The Uncounted Worlds

Muôn Thế Giới Chưa Đếm

Solutions to Chapter 1

Thirty Seconds on Four Worlds

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Contents

Chapter 1

Thirty Seconds on Four Worlds

1. How many geckos are there?

Mai, grade five. Biology and counting.

(a) At the very least: 48.

Step 1 41 geckos are already in the census; 26 were seen on Saturday. Adding those gives 67 — but 19 geckos are in both counts, and have been counted twice.

Step 2 Take the overlap off once:

$$41 + 26 - 19 \;=\; \boxed{48}$$

That is the inclusion–exclusion principle, and it rests on one assumption: that Mai recognises a gecko she has drawn. Given that, 48 different geckos have been seen, so at least 48 exist.

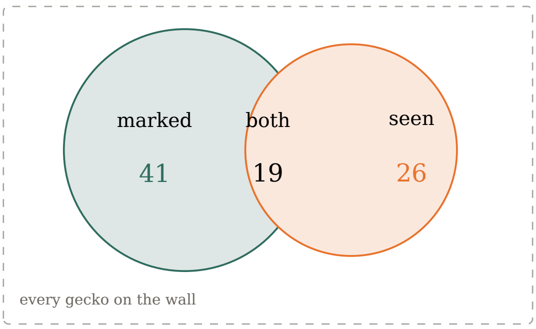
(b) Most likely: 56.

Step 1 Of the 26 seen on Saturday, 19 were already in the census: $\frac{19}{26} = 0.73\ldots$

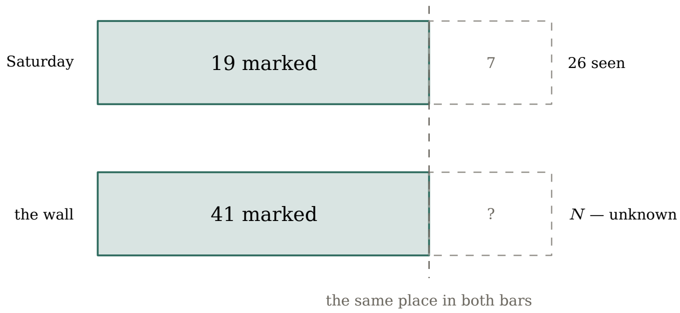
Step 2 If Saturday's geckos are a fair sample, the same fraction of the whole wall is already in the census.

Step 3 The number in the census is known exactly — 41 — so 41 is that same fraction of everything:

$$N \;\approx\; 41 \times \frac{26}{19} \;=\; \frac{1066}{19} \;\approx\; \boxed{56}$$



The overlap must not be counted twice.



The same fraction, in the sample and on the wall.

— ***The two answers are different kinds of number.*** *48 is a floor: certain, and no help at all in guessing how many are hiding. 56 is an estimate: it can be wrong, and it must never fall below the floor (48).*

— ***What the estimate assumes.*** *Three things.* Known and unknown geckos equally easy to see — *if the ones she already knows are easier to find, 19-out-of-26 is too high a fraction and the true total is larger.* No gecko in the census has died — *a dead one is still counted among the 41, which makes the estimate too large; hatchlings are harmless: a new gecko is not in the census, but it is one of the geckos Saturday's count is estimating.* Every gecko in the census is still recognisable — *one Mai fails to recognise moves out of the overlap, which also makes the estimate too large. (This is why she draws them rather than painting the skin.)*

For your toolkit: Capture–recapture.

$$\text{total} \;\approx\; \frac{(\text{number already known}) \times (\text{size of new sample})}{\text{already-known ones in the new sample}}$$

For counting a population you cannot see all at once: fish in a lake, words in a language, faults in a program. Say which way each assumption would bend the answer.

2. Why does it come back, and how far does it go?

Quang, grade seven. Geometry and physics.

(a) It travels 52.0 m — the same 52.0 m wherever he starts it.

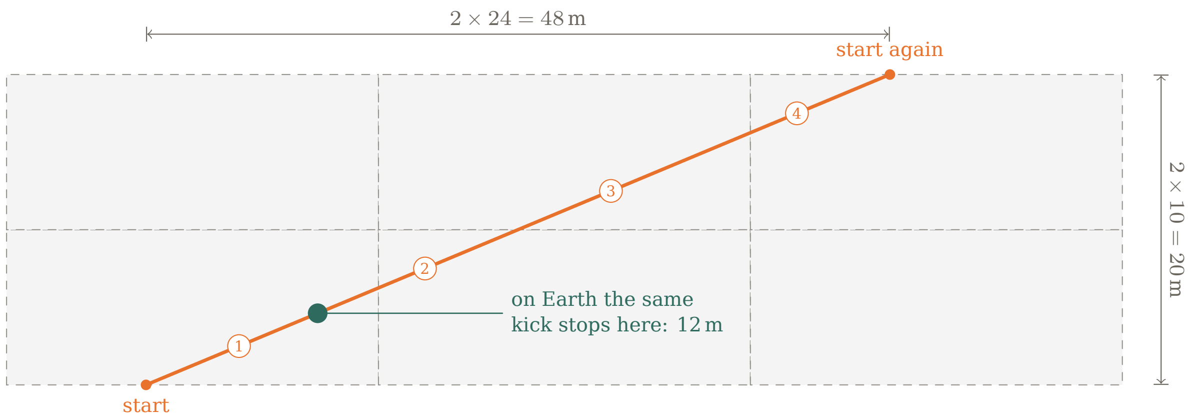
Step 1 Call the spot he sets the ball down O. A wall returns the ball at the angle it arrived, so rather than follow the ball round the corners, reflect the *court* at each wall and let the ball carry straight on into the mirror image. In the reflected copies the whole journey from O is one straight line.

Step 2 Kicked parallel to a diagonal, that straight line runs parallel to the diagonal of the reflected copies too. It comes back to a copy of its starting point after two court-lengths and two court-widths: $2 \times 24.0 = 48.0\,\text{m}$ across and $2 \times 10.0 = 20.0\,\text{m}$ up.

Step 3 Those are at right angles, so

$$\text{circuit} = \sqrt{48.0^2 + 20.0^2} = \sqrt{2704} = \boxed{52.0\text{ m}}$$

Step 4 On the way it crosses two vertical fold lines and two horizontal ones — one touch on each of the four walls — and O never enters the arithmetic. Move the start and the whole line slides sideways; both ends move together and its length does not change.



Reflect the court, not the ball, and the whole journey is one line.

— ***Why it stops in the same place, and not merely at the same distance.*** *Two facts have to meet. The journey is 52.0 m whatever point he starts from — that is the geometry. And friction takes the same speed out of the ball for every metre it rolls — that is the physics. Equal distance and equal loss per metre mean the ball runs out at the same point of its journey every time, which is the point it started from.*

— ***Of the paths that touch all four walls, the diagonal ones close soonest.*** *Kick parallel to a diagonal and the path shuts after four walls, one touch to a wall, in 52.0 m. Say that carefully. Shorter closed paths do exist: strike the ball square at a long wall and it runs there and back for ever, closing after 20.0 m and two touches — but it never meets the short walls at all. Among the paths that visit every wall, the diagonal ones are the shortest, because the unfolded line has to reach a copy of O that lies at least one court across and one court up. Most directions never close at all: the ball goes on bouncing and never returns. Which directions close, and after how many touches, is a longer question than it looks, and worth asking.*

(b) Not on Earth. The path would be identical; the ball would not get round it.

Step 1 Nothing in part (a) mentions gravity, or the ball. Reflections and Pythagoras do not care which world the court is on, so on Earth the circuit is still 52.0 m and still returns to O.

Step 2 What changes is how far the ball can roll. The ground pulls back with a friction force $\mu m g$, and that slows a ball of mass m at a rate

$$\frac{\mu m g}{m} = \mu g$$

The mass cancels: a heavy ball is harder to start and exactly as hard to stop, so it rolls no further than a light one from the same speed. Against friction, the weight of the ball is a red herring. Against the air it is not, and that is the next remark.

Step 3 Only g is left, and Earth's is $9.81/1.62 = 6.06$ times the Moon's, so the same kick dies in 6.06 times less distance. A pass that rolls 12.0 m on an Earth pitch — nothing in the question fixes that figure, it is here only to put a number on it — would carry $12.0 \times 6.06 \approx 73\,\mathrm{m}$ here — comfortably round a 52.0 m circuit.

Step 4 Turned round: to get round the same circuit at home, that pass would have to roll 52.0 m instead of 12.0 m. Distance goes as the square of the speed, so he would have to strike it $\sqrt{52.0/12.0} \approx \boxed{\text{2.1 times as fast}}$.

— ***The air was left out, and it is worth checking that it can be.*** *Old Dome is full of air, and the air drags on the ball as well as the ground does. Here the ball presses on the ground with a sixth of the weight it would have at home, so the ground's grip is weak and the air's share of the slowing is larger than it would be at home.*

You can see which way it goes without any new mathematics. Air pushes on the ball's front, so what the air can do depends on the ball's size; what resists being pushed is the ball's mass. The dome ball is a football's size and six times its mass, so the air shifts it about six times less. That is what it was built for — and notice what mass has just done. It fell out of the friction, and came straight back in through the air.

The numbers, for the curious. The dome ball gets round in about ***60 m*** *rather than 73: still home, but with 15% in hand instead of 40%. Kick an ordinary leather football instead — same size, a sixth of the mass — and it stops at about* ***36 m*** *and never gets home at all. (The roll is $\ln(1 + kv^2/\mu g)/2k$ with $k = \rho C_\mathrm{D} A/2m$; μ cancels out of the comparison, so none of this depends on the surface.)*

Check whether the term you dropped could change the answer, not merely whether it is small. This one nearly did — and it is the reason part (b) says an ordinary leather ball.

For your toolkit: Unfold the room, not the path.

For anything bouncing off straight walls at equal angles — a ball, a ray of light, a sound — reflect the room at each wall instead of turning the path. The journey straightens out and its length comes from one use of Pythagoras. In a rectangle $a \times b$, a path started parallel to a diagonal touches all four walls once and returns to its starting point after $2\sqrt{a^2 + b^2}$, wherever it started.

3. What are the two odd marks worth?

Linh, grade nine. Signal and language.

Five marks say their own value, so read them off:

ᛁ $= 0$ ᛧ $= 1$ ᚿ $= 2$ ᚼ $= 3$ ᚽ $= 4$

Write t for the triangle ᚹ and r for the ring ⚲, and b for the marks it takes to fill a column — the base. Biggest on the left, so ᛧ ᚼ is one full column and three over: $b + 3$.

(a) The triangle is 5, the ring is 6, and a column takes seven.

Step 1 Nothing is cut between the numbers, so guess the plainest thing — that a carving adds — and see what it costs. The first:

⚲ + ᚽ = ᛧ ᚼ $\qquad r + 4 = b + 3 \qquad r = b - 1$

Step 2 The second, on the same guess:

ᚹ + ⚲ = ᛧ ᚽ $\qquad t + (b - 1) = b + 4 \qquad t = 5$

The base cancels: the triangle is 5 whatever the base turns out to be. Two guesses, and one of them has already paid.

Step 3 Now the third, still as a sum:

ᚹ + ᚿ = ᛧ ᚼ $\qquad 5 + 2 = b + 3 \qquad b = 4$

which is impossible: that would make the triangle 5 in a system whose columns hold only 4, and no single mark can be worth as much as a full column. The guess fails here and only here, so it is this carving, not the other two, that is doing something else.

Step 4 There is one other thing two numbers can do. Take it as a product:

ᚹ × ᚿ = ᛧ ᚼ $\qquad 2 \times 5 = b + 3 \qquad \boxed{b = 7} \qquad$ ᚹ $= 5,$ ⚲ $= 6$

Every mark is now under seven, and all three carvings come out true.

(b) 2097.

Step 1 The row set apart is ⫯ | ⚑ ⫳ — 6, 0, 5, 4 in base seven. The bare stem is cut nowhere in the three carvings; it appears once, here, and says that this column holds nothing.

Step 2

$$6(343) + 0(49) + 5(7) + 4 \;=\; 2058 + 35 + 4 \;=\; \boxed{2097}$$

— ***Why that reading is the only one.*** *The challenge says each carving adds or multiplies, so there are eight ways to read the three together. Seven fail: they force a digit as large as its base, or a base that is not a whole number, or no base at all. Add, add, multiply is the only one left, and it works in base seven and in no other base. The last check is one nothing forced: the triangle and ring land exactly on 5 and 6, the two values the slants do not cover.*

That premise is Linh's. It is not the deck's, and it is doing more work than it looks.

Allow a third operation — a power, say — and other readings of the same three carvings become consistent too, in other bases, giving quite different numbers for the fourth row. Nothing cut into that deck rules them out. Nobody has been told they are wrong; they have only been set aside, which is a different thing.

So: find them. *Two of them are there, and the arithmetic is no harder than what you have just done — only the guessing is. Write down what you get and keep it somewhere. You will want it when you find out where Linh's assumption takes her.*

— ***The operation was never written, and never needed to be.*** *Plus and times are conveniences, not part of the arithmetic. Once you fix what the marks are worth and demand that every carving come out true, the arithmetic has only one way to be consistent, and that consistency names the operations for you. It is the same move an archaeologist makes with an unread tablet:* not *what does this sign mean,* but *which reading leaves nothing broken.*

— ***Seven marks, and then a new column.*** *Count what the system needs: nought, four slant-marks, the triangle, the ring. Seven signs, and the eighth number starts a fresh column. Ten fingers gave us ten digits, and wherever people have used another base you can still point at what they were counting on — twenty for fingers and toes, sixty out of twelve and five. Seven is not a hand.*

— ***Why the last two marks look different.*** *The first five say their own value, which is what you design when you expect to be read by someone who was never taught the system. Past four slants nobody can count at a glance, so the design changes shape rather than adding a fifth slant — the same reason we write* v *and not* iiii. *The two marks that gave Linh trouble are exactly where any counting system gives up.*

For your toolkit: Reading a number system you were not taught.

Say your assumptions out loud, then test them: a right set of guesses makes every statement true for exactly one reading, and a wrong set makes them true for none or for many. Turn each statement into an equation — some will pin a symbol without pinning the base, and those are the ones to do first. When a guess breaks, read the break: *a digit as big as its base* says the operation was guessed wrong, not the value. And remember that the operations are recoverable too. You are not owed the signs.

4. How far away did the light appear to come from?

Nam, grade eleven. Planetary science.

(a) About 18 km.

Step 1 In radians, $\theta = 0.15^\circ \times \frac{\pi}{180} = 2.618 \times 10^{-3}$.

Step 2 Rays from a single source, spreading across a baseline d at right angles to the source direction, open by $\theta \approx d/D$. So

$$D \;\approx\; \frac{d}{\theta} \;=\; \frac{48}{2.618 \times 10^{-3}} \;\approx\; \boxed{1.8 \times 10^4 \text{ m}} \;=\; 18 \text{ km}$$

(b) It cannot. Sunlight would fan by an angle half a million times finer than he can measure.

Step 1 Run the same formula backwards. The Sun is 1.5×10^{11} m away, so across the same 48 m baseline its rays open by

$$\theta \;=\; \frac{48}{1.5 \times 10^{11}} \;=\; 3.2 \times 10^{-10} \text{ rad} \;=\; 1.8 \times 10^{-8}\,{}^\circ$$

Step 2 His instrument resolves about $10^{-2}\,{}^\circ$ — $\boxed{\text{half a million times coarser}}$. No measurement he can make will ever show the Sun's rays fanning at all, and 0.15° is some seven orders of magnitude too large to be sunlight under any circumstances.

(c) About 275 km. Any further off, and these two masts could not have told it from the Sun.

Step 1 The shadows part only as far as the light fans, so the masts show anything at all only once θ reaches the instrument's floor, $10^{-2\,\circ} = 1.745 \times 10^{-4}$ rad.

Step 2 Put that floor into the same relation:

$$D = \frac{d}{\theta} = \frac{48}{1.745 \times 10^{-4}} \approx \boxed{2.8 \times 10^{5}\ \text{m}} = 275\ \text{km}$$

Step 3 Anything further away fans by less than he can measure, and two shadows forty-eight metres apart lie down parallel — which is exactly what they do under the Sun.

— ***275 km is not a distance to anything. It is the edge of what these masts could see.*** *Nam's 18 km sits well inside it. Had the same thing happened a thousand kilometres out, the eleven null days would have been twelve, and nobody would have known there was anything to know.*

— ***Eleven days of nothing was the correct reading, not a failure.*** *This is why the experiment had to run, and had to come out flat. An instrument that cannot resolve an effect will report its absence, correctly, for ever. The eleven null days are what make the twelfth measurement mean anything at all — without them,* $0.15°$ *is a glitch.*

— ***What the 18 km rests on.*** *The measurement establishes only that the light was not parallel across 48 m. The distance follows if there was one source, if it was small compared with 18 km, and if the light travelled in straight lines. None of the three has been checked. None of the three affects the finding that it was not sunlight.*

For your toolkit: Small-angle parallax.

Parallax is the shift in the apparent direction of a thing when it is seen from two different places; the nearer it is, the bigger the shift. For small θ, $\theta \approx d/D$ in radians: a known baseline — the distance between the two viewpoints — and a measured angle give a distance. Parallel rays mean a far source, fanning rays a near one.